A Multi-Spectroscopic Approach to Monitor Catalyst Deactivation in Homogeneous Catalysis

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To improve the homogenously catalyzed **hydroformylation reaction** by ensuring constant activity and selectivity during continuous processes, emphasis must be placed on the **catalytic stability that has been** insufficiently studied. Catalyst deactivation – having many potential causes – leads to undesired byproducts, since changes in the **molecular configuration** of the catalyst induce alternative competing reaction pathways. Integration of renewables with inconsistent quality into this process further signifies the need to understand the exact mechanisms behind **ligand degradation and catalyst deactivation**. The catalyst deactivation modes, which will be studied, are presented in the Fig. 1.

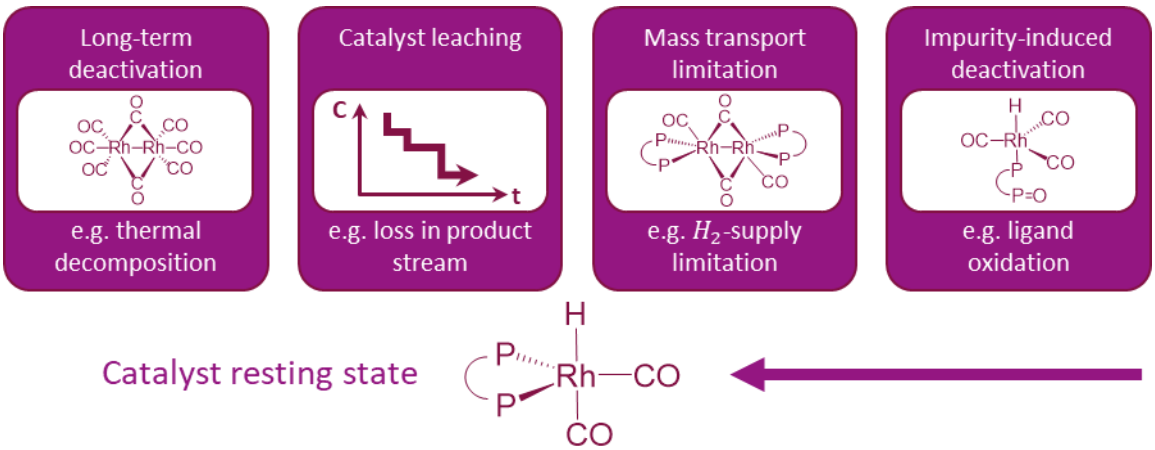

Fig. 1: Catalyst deactivation modes in the hydroformylation.

To overcome the data limitations that often obscure the underlying chemistry in complex systems a **multi-spectroscopic approach** under industrially relevant conditions (100°C, 30 bar) will be employed for the hydroformylation of 1-decene using

modified rhodium catalysts ($Rh(acac)(CO)_2$ with various ligands). To detect and quantify reaction and catalytic components through a combination of complementary data this strategy aims to integrate:

(1) **FTIR** flow cell to detect rhodium-carbonyl vibrational frequencies;

(2) **Raman** spectrometer (532.0 and 623.8 nm) with immersion optics to detect vibrational signatures invisible or complementary to the FTIR;

(3) **NMR** (benchtop, ^{31}P, ^{1}H) with a high-pressure flow cell for monitoring phosphor containing ligands and catalyst integrity over time;

(4) **UV–Vis** setup to observe electronic transitions linked to metal-ligand interactions.

The setup developed and used can be seen in the Fig. 2.

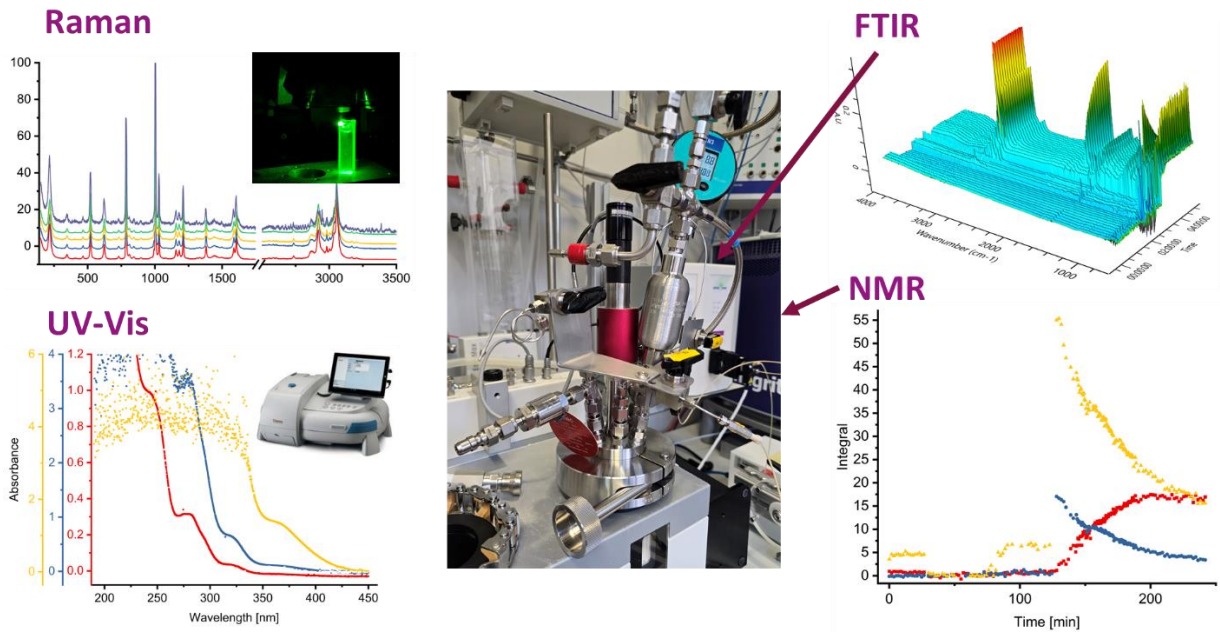

Fig. 2: Reactor setup and results from multi-spectroscopy.

This integrated multi-spectroscopic methodology is expected to yield a robust, data-driven foundation for **interpreting catalyst deactivation pathways** and devising **stabilization strategies** that reduce byproduct formation and ultimately enhance selectivity, longevity, and sustainability.

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